

Digital Intelligent Turbine Flowmeter Communication Protocol

Type E

1. Introduction

This instrument adopts the MODBUS_RTU format.

The default data format is n,8,1 (1 start bit, 8 data bits, no parity, 1 stop bit), supporting options such as odd/even parity and 2 stop bits.

The default baud rate is 9600, with options of 1200, 2400, 4800, 9600, and 19200.

The instrument address ranges from decimal "01" to "255", with address "0" reserved for broadcast.

This protocol does not support broadcasting, and all other addresses are reserved. This instrument utilizes the MODBUS protocol's 0x03 command.

Command 03(HEX)	Read Single or Multiple Registers
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Protocol Data Types:

Single-precision floating-point number(float), formatted as IEEE754, with data ordered from high to low.

Double-precision floating-point number (double), formatted as IEEE754, with data ordered from high to low.

Unsigned integer (unsigned int) ranging from 0 to 65535.

Command 3 format (read register command):

MODBUS request

Instrument Address	1 BYTE	01-FF
Function Code	1 BYTE	03
Starting Address	2 BYTE	0-FFFF
Number of Readings	2 BYTE	N(1-7D)
CRC Low Byte	1 BYTE	
CRC High Byte	1 BYTE	

MODBUS response

Instrument Address	1 BYTE	01-FF
Function Code	1 BYTE	03
Byte Count	1 BYTE	N*2
Input Status	N*2 BYTE	
CRC Low Byte	1 BYTE	
CRC High Byte	1 BYTE	

Error Response

Instrument Address	1 BYTE	01-FF
Function Code	1 BYTE	83
Error Code	1 BYTE	01, 02, 03(Refer to 1)
CRC Low Byte	1 BYTE	
CRC High Byte	1 BYTE	

Note 1:

01 - Register Address Error

02 - Register Length Error

03 - CRC Error

2. Data Item Definitions

Attribute	Name	Address (Hexadecimal)	Register Length (Words)	Data Type	Description
Read-only	Cumulative Flow	0000	4	Double float	Cumulative Flow in Double Precision Floating Point Format
Read-only	Cumulative Flow	0004	2	float	Cumulative Flow in Single Precision Floating Point Format
Read-only	Instantaneous Flow	0006	2	float	
Read-only	Instantaneous Flow Unit	0008	1	unsigned int	0:m3/h 1:L/h 2:L/min 3:US Gal/min 4:UK Gal/min 5:US Gal/h 6:UK Gal/h 7:kg/h 8:t/h 9:ft3/h
Read-only	Battery Voltage	0009	2	float	Unit, V

Communication Example: (Instrument Address is 01)

Read Command	01 03 0004 0004 05C8	CRC Low Byte First, Read Cumulative Flow and Instantaneous Flow Simultaneously
Returned Data	01 03 08 42 84 00 00(Cumulative Flow=66) 00 00 00 00(Instantaneous Flow=0) D4 36	Floating-point numbers: High byte first, CRC low byte first